

浙大量子 2022.

一、质量为  $\mu$  的粒子处于一维谐振子势中，在初始时刻的波函数为  $\psi(x, t=0) = A(2\varphi_0 + \varphi_1)$

(1) 求归一化系数  $A$

(2) 求  $t > 0$  时刻的波函数  $\psi(x, t)$

(3) 求  $t = T$  时刻，体系的能量。

(4) 求  $x$  的平均值。

(1) 由  $\int |\psi(x, t=0)|^2 dx = 1$  得

$$|A|^2 \int (2\varphi_0^* + \varphi_1^*)(2\varphi_0 + \varphi_1) dx = 1$$

$$\Rightarrow |A|^2 [4 \int |\varphi_0|^2 dx + \int |\varphi_1|^2 dx] = 5|A|^2 = 1$$

$$A = \frac{1}{\sqrt{5}}$$

(2) 一维谐振子势中， $E_n = (n + \frac{1}{2})\hbar\omega$

$$\psi(x, t) = \frac{2}{\sqrt{5}} e^{-iE_0 t/\hbar} \varphi_0 + \frac{1}{\sqrt{5}} e^{-iE_1 t/\hbar} \varphi_1$$

$$= \frac{2}{\sqrt{5}} e^{-i\frac{1}{2}\omega t} \varphi_0 + \frac{1}{\sqrt{5}} e^{-i\frac{3}{2}\omega t} \varphi_1$$

$$(3) E = \langle \psi(x, T) | \hat{H} | \psi(x, T) \rangle$$

$$= \frac{4}{5} \langle \varphi_0 | \hat{H} | \varphi_0 \rangle + \frac{1}{5} \langle \varphi_1 | \hat{H} | \varphi_1 \rangle$$

$$= \frac{4}{5} \times \frac{1}{2} \hbar\omega + \frac{1}{5} \times \frac{3}{2} \hbar\omega$$

$$= \frac{7}{10} \hbar\omega$$

$$(4) x \varphi_n = \frac{1}{2} \left[ \sqrt{\frac{n}{2}} \varphi_{n-1} + \sqrt{\frac{n+1}{2}} \varphi_{n+1} \right]$$

$$\langle x \rangle = \langle \psi(x, t) | x | \psi(x, t) \rangle$$

$$= \frac{1}{5} \int (2\varphi_0^* e^{i\frac{1}{2}\omega t} + \varphi_1^* e^{i\frac{3}{2}\omega t})$$

$$x (2\varphi_0 e^{-i\frac{1}{2}\omega t} + \varphi_1 e^{-i\frac{3}{2}\omega t}) dx$$

$$= \frac{1}{5} \int (2\varphi_0^* e^{i\frac{1}{2}\omega t} + \varphi_1^* e^{i\frac{3}{2}\omega t})$$

$$\frac{1}{2} [\sqrt{2} \varphi_1 e^{-i\frac{1}{2}\omega t} + \frac{1}{\sqrt{2}} \varphi_0 e^{-i\frac{3}{2}\omega t} + \varphi_2 e^{-i\frac{5}{2}\omega t}] dx$$

$$= \frac{1}{5} \int [\sqrt{2} |\varphi_0|^2 e^{-i\omega t} + \sqrt{2} |\varphi_1|^2 e^{i\omega t}] dx$$

$$= \frac{2\sqrt{2}}{5} \cos \omega t$$

二、质量为  $\mu$  的粒子处于  $0 \leq x \leq a$  的一维无限深

势阱中，在初始时刻的波函数为

$$\psi(x, t=0) = A(1 + \cos \frac{\pi x}{a}) \sin \frac{\pi x}{a}$$

(1) 求归一化系数  $A$

(2) 求  $t = T$  时刻，体系的能量

(3) 求  $t > 0$  时刻的波函数  $\psi(x, t)$

(4) 求在  $t = 0$  时刻，粒子处于  $0 \leq x \leq \frac{a}{2}$  的概率。

(1) 一维无限深势阱

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

$$\psi(x, t=0) = A \sin \frac{\pi x}{a} + A \cos \frac{\pi x}{a} \sin \frac{\pi x}{a}$$

$$= A \sqrt{\frac{a}{2}} \psi_1 + \frac{A}{2} \sqrt{\frac{a}{2}} \psi_2$$

由归一化  $\int |\psi(x, t=0)|^2 dx = 1$  得

$$|A \sqrt{\frac{a}{2}}|^2 + |\frac{A}{2} \sqrt{\frac{a}{2}}|^2 = 1$$

$$A = \sqrt{\frac{8}{5a}}$$

$$\psi(x, t=0) = \sqrt{\frac{4}{5}} \psi_1 + \sqrt{\frac{1}{5}} \psi_2$$

(2) 体系能量不随时间改变。

$$t = T \text{ 时, } E = \frac{4}{5} E_1 + \frac{1}{5} E_2$$

$$= \frac{4\pi^2 \hbar^2}{5ma^2}$$

$$(3) t > 0 \text{ 时, } \psi(x, t) = \sqrt{\frac{4}{5}} e^{-iE_1 t/\hbar} \psi_1 + \sqrt{\frac{1}{5}} e^{-iE_2 t/\hbar} \psi_2$$

$$= \sqrt{\frac{4}{5}} e^{-i \frac{\pi^2 \hbar t}{2ma^2}} \psi_1 + \sqrt{\frac{1}{5}} e^{-i \frac{2\pi^2 \hbar t}{ma^2}} \psi_2$$

$$= \sqrt{\frac{8}{5a}} e^{-i \frac{\pi^2 \hbar t}{2ma^2}} \sin \frac{\pi x}{a} + \sqrt{\frac{2}{5a}} e^{-i \frac{2\pi^2 \hbar t}{ma^2}} \sin \frac{2\pi x}{a}$$

(4) 概率为

$$\int_0^{\frac{a}{2}} |\psi(x, t=0)|^2 dx$$

$$= \int_0^{\frac{a}{2}} \frac{8}{5a} (1 + \cos \frac{\pi x}{a})^2 \sin^2 \frac{\pi x}{a} dx$$

$$= \frac{8}{5\pi} \int_0^{\frac{\pi}{2}} (1 + \cos t)^2 \sin^2 t dt$$

$$= \frac{8}{5\pi} \int_0^{\frac{\pi}{2}} (\sin^2 t + 2 \sin^2 t \cos t + \sin^2 t \cos^2 t) dt$$

$$= \frac{8}{5\pi} \left( \frac{5\pi}{16} + \frac{2}{3} \right)$$

$$= \frac{1}{2} + \frac{16}{15\pi} \quad \text{X} \quad t_0 \text{ 时刻}$$

$$\int_0^{\frac{a}{2}} |\psi(x, t_0)|^2 dx$$

$$\text{令 } \frac{\pi^2 \hbar}{2ma^2} = \lambda$$

$$\psi(x, t_0) = \sqrt{\frac{4}{5}} e^{-i\lambda t_0} \psi_1 + \sqrt{\frac{1}{5}} e^{-i4\lambda t_0} \psi_2$$

$$\int_0^{\frac{a}{2}} |\psi(x, t_0)|^2 dx$$

$$= \int_0^{\frac{a}{2}} \frac{4}{5} |\psi_1|^2 + \frac{1}{5} |\psi_2|^2 + \frac{2}{5} \psi_1 \psi_2 (e^{-i3\lambda t_0} + e^{i3\lambda t_0}) dx$$

$$= \frac{1}{2} + \int_0^{\frac{a}{2}} \frac{4}{5} \psi_1 \psi_2 \cos 3\lambda t_0 dx$$

$$= \frac{1}{2} + \frac{16}{5a} \cos 3\lambda t_0 \int_0^{\frac{a}{2}} \sin^2 \frac{\pi x}{a} \cos \frac{\pi x}{a} dx$$

$$= \frac{1}{2} + \frac{16}{5a} \cos 3\lambda t_0 \cdot \frac{a}{3\pi}$$

$$= \frac{1}{2} + \frac{16}{15\pi} \cos \frac{3\pi^2 \hbar t_0}{2ma^2}$$

三、有一质量为  $m$  的粒子在长度为  $L$  的圆环上自由运动。

(1) 求定态波函数和能级。

(2) 在  $t=0$  时刻，体系的初始波函数为

$\psi(\varphi, t=0) = A \sin^2 \varphi$ ，求  $t>0$  时刻的波函数  $\psi(\varphi, t)$

(3) 若体系受到微扰  $H' = \lambda \sin \varphi \cos \varphi$  的作用，

其中  $\lambda$  为常数，计算能级的一级修正。

$$(1) \hat{L}_z = -i\hbar \frac{d}{d\varphi}$$

$$\hat{H} = \frac{\hat{L}_z^2}{2I} = -\frac{\hbar^2}{2I} \frac{d^2}{d\varphi^2}$$

$$\text{由 } \hat{H}\psi(\varphi) = E\psi(\varphi)$$

$$\Rightarrow \frac{d^2}{d\varphi^2} \psi(\varphi) + \frac{2IE}{\hbar^2} \psi(\varphi) = 0$$

$$\text{令 } m = \sqrt{\frac{2IE}{\hbar^2}}$$

$$\psi(\varphi) = Ae^{im\varphi}$$

$$\text{由边界条件 } \psi(2\pi) = \psi(0)$$

$$\Rightarrow e^{i2\pi m} = 1$$

$$m = 0, \pm 1, \pm 2, \dots$$

$$\text{归一化, } \psi_m(\varphi) = \frac{1}{\sqrt{2\pi}} e^{im\varphi}, m=0, \pm 1, \pm 2, \dots$$

$$E_m = \frac{m^2 \hbar^2}{2I}$$

$$(2) \psi(\varphi, t=0) = A \sin^2 \varphi = \frac{A}{2} (1 - \cos 2\varphi)$$

$$= \frac{A}{2} \left[ 1 - \frac{1}{2} (e^{i2\varphi} + e^{-i2\varphi}) \right]$$

$$= \sqrt{\pi} \left[ \frac{A}{2} \psi_0(\varphi) - \frac{A}{4} \psi_2(\varphi) - \frac{A}{4} \psi_{-2}(\varphi) \right]$$

$$\text{归一化: } \int |\psi(\varphi, t=0)|^2 d\varphi = 1$$

$$\Rightarrow 2\pi \left( \frac{|A|^2}{4} + \frac{|A|^2}{16} + \frac{|A|^2}{16} \right) = \frac{3\pi}{4} |A|^2 = 1 \Rightarrow A = \sqrt{\frac{4}{3\pi}}$$

$$\psi(\varphi, t=0) = \sqrt{\frac{2}{3}} \psi_0 - \sqrt{\frac{1}{6}} \psi_2 - \sqrt{\frac{1}{6}} \psi_{-2}$$

$$\psi(\varphi, t) = \sqrt{\frac{2}{3}} e^{-iE_0 t/\hbar} \psi_0 - \sqrt{\frac{1}{6}} e^{-iE_2 t/\hbar} \psi_2 - \sqrt{\frac{1}{6}} e^{-iE_{-2} t/\hbar} \psi_{-2}$$

$$= \sqrt{\frac{2}{3}} \psi_0 - \sqrt{\frac{1}{6}} e^{-i2\hbar t/I} \psi_2 - \sqrt{\frac{1}{6}} e^{-i2\hbar t/I} \psi_{-2}$$

$$= \sqrt{\frac{1}{3\pi}} - \sqrt{\frac{1}{12\pi}} e^{-i2\hbar t/I} (e^{i2\varphi} + e^{-i2\varphi})$$

$$= \sqrt{\frac{1}{3\pi}} - \sqrt{\frac{1}{3\pi}} \cos 2\varphi e^{-i2\hbar t/I}$$

(3)  $m=0$  时，非简并。

$m \neq 0$  时， $E_m = E_{-m}$ ，二级简并。

$$m=0, \psi_0 = \frac{1}{\sqrt{2\pi}}$$

$$E_0^{(1)} = \langle \psi_0 | H' | \psi_0 \rangle$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \lambda \sin \varphi \cos \varphi d\varphi = 0$$

$m \neq 0$  时，

$$|1\rangle = \psi_m(\varphi) = \frac{1}{\sqrt{2\pi}} e^{im\varphi}$$

$$|2\rangle = \psi_{-m}(\varphi) = \frac{1}{\sqrt{2\pi}} e^{-im\varphi}$$

$$H'_{11} = \langle 1 | H' | 1 \rangle$$

$$= \frac{1}{2\pi} \int_0^{2\pi} e^{-im\varphi} \lambda \sin \varphi \cos \varphi e^{im\varphi} d\varphi = 0$$

$$H'_{12} = \langle 1 | H' | 2 \rangle$$

$$= \frac{1}{2\pi} \int_0^{2\pi} e^{-im\varphi} \lambda \sin \varphi \cos \varphi e^{-im\varphi} d\varphi$$

$$= \frac{1}{2\pi} \int_0^{2\pi} e^{-i2m\varphi} \cdot \frac{\lambda}{4i} (e^{i2\varphi} - e^{-i2\varphi}) d\varphi$$

$$= \frac{i\lambda}{8\pi} (\delta_{m,1} - \delta_{m,-1})$$

$$H'_{21} = H'_{12}^*$$

$$H'_{22} = \langle 2 | H' | 2 \rangle = 0$$

当  $|m| > 1$  时,  $E_m^{(1)} = 0$ .

$m=1$  与  $m=-1$  时等价, 考虑  $m=1$ .

$$H' = \begin{pmatrix} 0 & \frac{i\lambda}{8\pi} \\ -\frac{i\lambda}{8\pi} & 0 \end{pmatrix}$$

$$\text{由 } \begin{vmatrix} E^{(1)} & -\frac{i\lambda}{8\pi} \\ \frac{i\lambda}{8\pi} & E^{(1)} \end{vmatrix} = 0$$

$$\Rightarrow E_1^{(1)} = \pm \frac{\lambda}{8\pi}$$

四、

(2) 跃迁公式

$$W_{(100) \rightarrow (21m)}(t) = \frac{1}{\hbar^2} \left| \int_0^t H'_{(21m)(100)}(t) e^{i\omega_{21}t} dt \right|^2$$

$$\omega_{21} = \frac{E_2 - E_1}{\hbar} = \frac{3e^2}{8a\hbar}$$

$$H'_{(21\pm 1)(100)} = 0$$

$$H'_{(210)(100)} = \frac{2^7 \sqrt{2} e \epsilon_0 a e^{-t/\tau}}{3^5}$$

$$W_{(100) \rightarrow (21\pm 1)}(t) = 0$$

$$W_{(100) \rightarrow (210)}(t)$$

$$= \frac{1}{\hbar^2} \left| \int_0^t \frac{2^7 \sqrt{2} e \epsilon_0 a}{3^5} e^{-t/\tau} e^{i\omega_{21}t} dt \right|^2$$

$$= \frac{2^{15} e^2 \epsilon_0^2 a^2}{3^{10} \hbar^2} \left| \int_0^t e^{-t/\tau} e^{i\omega_{21}t} dt \right|^2$$

$$= \frac{2^{15} e^2 \epsilon_0^2 a^2}{3^{10} \hbar^2} \frac{1}{(1/\tau)^2 + (\omega_{21})^2}$$

四、原处于基态的氢原子, 受到一在方向一致、空间均匀、但随时间衰减的电场的影响

$$\epsilon(t) = \begin{cases} 0, & t < 0 \\ \epsilon_0 e^{-t/\tau}, & t > 0 \end{cases}$$

其中  $\epsilon_0, \tau > 0$  为常数. 取电场方向为  $z$  轴方向, 电场对氢原子的影响视为微扰  $H' = -q\vec{r} \cdot \vec{E}$ .

已知

$$\psi_{100} = \sqrt{\frac{1}{\pi a^3}} e^{-r/a}, \quad \psi_{210} = \frac{1}{4a^2} \sqrt{\frac{1}{2\pi a}} r e^{-r/2a} \cos\theta$$

$$\psi_{21\pm 1} = \mp \frac{1}{8a^2} \sqrt{\frac{1}{\pi a}} r e^{-r/2a} \sin\theta e^{\pm i\varphi}$$

(1) 求  $\langle 21m | H' | 100 \rangle$  的矩阵元.

(2) 当  $t \gg \tau$  时, 求基态氢原子跃迁到  $2p$  的概率.

$$H' = -q\vec{r} \cdot \vec{E} = \begin{cases} 0, & t < 0 \\ -q\epsilon_0 e^{-t/\tau} z, & t > 0 \end{cases}$$

$t < 0$  时矩阵元都为 0

$t > 0$  时,

$$\langle 210 | H' | 100 \rangle = -q\epsilon_0 e^{-t/\tau} \langle 210 | z | 100 \rangle$$

$$= -q\epsilon_0 e^{-t/\tau} \int \frac{1}{4a^2} \sqrt{\frac{1}{2\pi a}} r e^{-r/2a} \cos\theta (r \cos\theta) \sqrt{\frac{1}{\pi a^3}} e^{-r/a} r^2 \sin\theta dr d\theta d\varphi$$

$$= -q\epsilon_0 e^{-t/\tau} \frac{1}{4\sqrt{2}\pi a^4} \int_0^\infty r^4 e^{-3r/2a} dr \int_0^\pi \cos^2\theta \sin\theta d\theta \int_0^{2\pi} d\varphi$$

$$= -q\epsilon_0 e^{-t/\tau} \frac{1}{4\sqrt{2}\pi a^4} \left( \frac{a}{2} \right)^5 \cdot 24 \cdot \frac{2}{3} \cdot 2\pi$$

$$= -\frac{q\epsilon_0 a e^{-t/\tau}}{4\sqrt{2}} \frac{2^7 \sqrt{2} e \epsilon_0 a e^{-t/\tau}}{3^5}$$

$$\langle 21\pm 1 | H' | 100 \rangle = -q\epsilon_0 e^{-t/\tau} \langle 21\pm 1 | z | 100 \rangle$$

$$= -q\epsilon_0 e^{-t/\tau} \int \left( \mp \frac{1}{8a^2} \sqrt{\frac{1}{\pi a}} r e^{-r/2a} \sin\theta e^{\pm i\varphi} \right) (r \cos\theta) \sqrt{\frac{1}{\pi a^3}} e^{-r/a} r^2 \sin\theta d\tau$$

$$= A \int_0^{2\pi} e^{\pm i\varphi} d\varphi = 0$$

五: 在均匀磁场中, 两电荷异号自旋为  $\frac{1}{2}$  的粒子的哈密顿量  $H = H_1 + H_p$ , 其中  $H_1 = -a \hat{S}_1 \cdot \hat{S}_2$ ,  $H_p = -\frac{b}{4\hbar} \hat{S}_{1z} + \frac{b}{4\hbar} \hat{S}_{2z}$ . 试在  $\hat{S}^2, \hat{S}_z$  表象中  
 (1) 求  $H$  的本征态和本征值.

设  $|1\rangle = \alpha(1)\alpha(2)$ ,  $|2\rangle = \alpha(1)\beta(2)$

$|3\rangle = \beta(1)\alpha(2)$ ,  $|4\rangle = \beta(1)\beta(2)$

$\hat{H}_1 = -a(\hat{S}_{1x}\hat{S}_{2x} + \hat{S}_{1y}\hat{S}_{2y} + \hat{S}_{1z}\hat{S}_{2z})$

$\hat{H}_1|1\rangle = -a[\hat{S}_{1x}\alpha(1)\hat{S}_{2x}\alpha(2) + \hat{S}_{1y}\alpha(1)\hat{S}_{2y}\alpha(2) + \hat{S}_{1z}\alpha(1)\hat{S}_{2z}\alpha(2)]$

$= -\frac{\hbar^2 a}{4}[\beta(1)\beta(2) + i\beta(1)i\beta(2) + \alpha(1)\alpha(2)]$

$= -\frac{\hbar^2 a}{4}\alpha(1)\alpha(2) = -\frac{\hbar^2 a}{4}|1\rangle$

$\hat{H}_p|1\rangle = -\frac{b}{4\hbar}\hat{S}_{1z}\alpha(1)\alpha(2) + \frac{b}{4\hbar}\alpha(1)\hat{S}_{2z}\alpha(2)$

$= -\frac{b}{8}\alpha(1)\alpha(2) + \frac{b}{8}\alpha(1)\alpha(2) = 0$

$\hat{H}_1|2\rangle = -a[\hat{S}_{1x}\alpha(1)\hat{S}_{2x}\beta(2) + \hat{S}_{1y}\alpha(1)\hat{S}_{2y}\beta(2) + \hat{S}_{1z}\alpha(1)\hat{S}_{2z}\beta(2)]$

$= -a\frac{\hbar^2}{4}[\beta(1)\alpha(2) + i\beta(1)\{-i\alpha(2)\} - \alpha(1)\beta(2)]$

$= -\frac{\hbar^2 a}{4}[2\beta(1)\alpha(2) - \alpha(1)\beta(2)]$

$= -\frac{\hbar^2 a}{2}|3\rangle + \frac{\hbar^2 a}{4}|2\rangle$

$\hat{H}_p|2\rangle = -\frac{b}{4\hbar}\hat{S}_{1z}\alpha(1)\beta(2) + \frac{b}{4\hbar}\alpha(1)\hat{S}_{2z}\beta(2)$

$= -\frac{b}{4}\alpha(1)\beta(2) = -\frac{b}{4}|2\rangle$

$\hat{H}_1|3\rangle = -\frac{\hbar^2 a}{4}[\alpha(1)\beta(2) - i\alpha(1)i\beta(2) - \beta(1)\alpha(2)]$

$= -\frac{\hbar^2 a}{2}|2\rangle + \frac{\hbar^2 a}{4}|3\rangle$

$\hat{H}_p|3\rangle = \frac{b}{4}\beta(1)\alpha(2) = \frac{b}{4}|3\rangle$

$\hat{H}_1|4\rangle = -\frac{\hbar^2 a}{4}[\alpha(1)\alpha(2) - \alpha(1)\alpha(2) + \beta(1)\beta(2)]$   
 $= -\frac{\hbar^2 a}{4}|4\rangle$

$\hat{H}_p|4\rangle = \frac{b}{8}\beta(1)\beta(2) - \frac{b}{8}\beta(1)\beta(2) = 0$

$H$  的矩阵为

$$H = \begin{pmatrix} -\frac{\hbar^2 a}{4} & 0 & 0 & 0 \\ 0 & \frac{\hbar^2 a}{4} - \frac{b}{4} & -\frac{\hbar^2 a}{2} & 0 \\ 0 & -\frac{\hbar^2 a}{2} & \frac{\hbar^2 a}{4} + \frac{b}{4} & 0 \\ 0 & 0 & 0 & -\frac{\hbar^2 a}{4} \end{pmatrix}$$

可以看出  $|1\rangle$  和  $|4\rangle$  是本征值  $E_1 = E_4 = -\frac{\hbar^2 a}{4}$  的本征态.

在  $|2\rangle$  和  $|3\rangle$  为基的空间中.

$$H = \begin{pmatrix} \frac{\hbar^2 a}{4} - \frac{b}{4} & -\frac{\hbar^2 a}{2} \\ -\frac{\hbar^2 a}{2} & \frac{\hbar^2 a}{4} + \frac{b}{4} \end{pmatrix}$$

令  $A = \frac{\hbar^2 a}{4}$ ,  $B = \frac{b}{4}$ .

由  $\begin{pmatrix} A-B & -2A \\ -2A & A+B \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = E \begin{pmatrix} x \\ y \end{pmatrix}$

$\Rightarrow \begin{vmatrix} E-A+B & 2A \\ 2A & E-A-B \end{vmatrix} = 0$

$E = A \pm \sqrt{B^2 + 4A^2}$

$= \frac{\hbar^2 a}{4} \pm \frac{1}{4}\sqrt{b^2 + 4\hbar^4 a^2}$

令  $\sqrt{B^2 + 4A^2} = C$ .

当  $E = A + C$  时,  $\psi = \frac{1}{\sqrt{4A^2 + (B+C)^2}} \begin{pmatrix} 2A \\ B+C \end{pmatrix}$

$= \frac{1}{\sqrt{4A^2 + (B+C)^2}} (2A|2\rangle + (B+C)|3\rangle)$

$E = A - C$  时,  $\psi = \frac{1}{\sqrt{4A^2 + (B-C)^2}} \begin{pmatrix} 2A \\ B-C \end{pmatrix}$

$= \frac{1}{\sqrt{4A^2 + (B-C)^2}} (2A|2\rangle + (B-C)|3\rangle)$